GAKD: Geometry-Aware Knowledge Distillation for Peer-to-Peer Federated Learning

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***Abstract*—Peer-to-peer federated learning enables decentralized model training without relying on a central server by allowing clients to sequentially exchange and update a shared model. FedPC formulates this process as continual learning across clients and mitigates catastrophic forgetting through proximal regularization in the parameter space. However, preserving proximity in the parameter space does not necessarily preserve the underlying geometry of the representation, which is crucial for knowledge transfer, especially under highly non-IID data distributions across clients. This limitation becomes particularly evident when considering new client compatibility, where models must rapidly adapt to previously unseen clients within a limited number of gradient update steps. In this work, we propose a representation-preserving learning objective for peer-to-peer federated learning that replaces parameter-space regularization with a combination of supervised contrastive learning and relational knowledge distillation, explicitly improving cross-client generalization and new client compatibility under strong non-IID data distributions.**

***Index Terms*—Peer-to-peer federated learning, Continual learning, Contrastive learning, Knowledge distillation**

I. Introduction

A. Motivation and Background

Vision-based sensing systems have become an integral component of modern driver monitoring applications, including driver activity recognition and advanced driver assistance technologies. Naturalistic Driving Action Recognition (NDAR) [1] is one such application aimed at enhancing road safety and reducing traffic accidents. In NDAR, monitoring systems learn patterns of human driving behavior to detect driver distractions, recognize driver activities, and predict vehicle trajectories. These systems rely on large-scale data collected from diverse users and driving environments. However, such data may contain sensitive information, including personal habits and location traces.

Federated Learning (FL) [2] has emerged as a promising paradigm for addressing these challenges by enabling collaborative model training in distributed environments without requiring raw data to leave client devices. Instead of aggregating data at a central location, FL coordinates the exchange and aggregation of model updates across participating clients. This decentralized training framework has been widely adopted in privacy-sensitive applications where preserving data confidentiality is critical.

Most existing FL frameworks, however, rely on a central server to orchestrate the training process and aggregate model updates from participating clients, a paradigm commonly referred to as client-to-server federated learning (C2S-FL). This centralized paradigm introduces several limitations, including a single point of failure, potential communication bottlenecks at the server, and a strong dependence on centralized trust assumptions. To address these challenges, peer-to-peer federated learning (P2P-FL) has emerged, enabling decentralized model updates through direct client-to-client exchange without reliance on a central server.

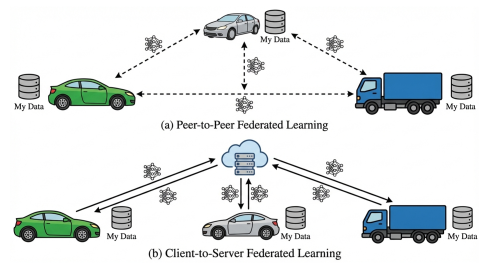

Fig. 1. **Comparison of P2P-FL and C2S-FL.** In P2P-FL, models are exchanged directly among clients without a central server, whereas in C2S-FL, a central server aggregates and redistributes model updates.

B. Challenges of Heterogeneous Data in Federated Learning

Data heterogeneity is a fundamental challenge in FL, including P2P-FL. In NDAR, this issue is particularly pronounced, as each driver exhibits unique behavioral patterns and driving styles and operates under diverse environmental conditions. This results in highly non-independent and non-identically distributed (non-IID) data distributions across participating clients.

In P2P-FL, existing approaches address non-IID data distributions through mutual refinement strategies [3], [4]. However, these methods do not explicitly consider the interplay between non-IID data distributions and catastrophic forgetting, where sequential peer-to-peer model updates over non-IID data distributions can progressively overwrite previously learned knowledge. This effect is particularly pronounced in P2P-FL due to the absence of a central coordinator and the continual drift induced by diverse non-IID data distributions across clients. To the best of our knowledge, FedPC [5] is among the few methods that explicitly address catastrophic forgetting in P2P-FL. By formulating P2P-FL as a continual learning problem, FedPC captures the degradation of previously acquired knowledge caused by sequential updates on non-IID data distributions. To mitigate this issue, it introduces a proximal regularization term in parameter space that penalizes large deviations from the previous client model.

C. Representation Drift: A Core Challenge

While the FedPC objective constrains parameter drift, it does not explicitly regulate the evolution of feature representations. In deep neural networks, task-relevant information is encoded in the geometry of the representation space; thus, even small parameter updates can induce substantial representational shifts. This issue is amplified in P2P-FL, where sequential client updates accumulate over exchanges. Consequently, despite parameter-space regularization, transferable representations may degrade, limiting generalization.

D. Knowledge Distillation for Representation Preservation

Knowledge distillation (KD) has been widely adopted to improve knowledge transfer in FL; however, its effectiveness in mitigating representation drift remains limited. Existing KD approaches, such as parameter-space regularization [5] and logit-based distillation [6], constrain model parameters or outputs but do not explicitly preserve the underlying representation geometry. As a result, representation drift can accumulate during sequential client updates under non-IID data distributions and heterogeneous client conditions (e.g., differences in user behavior and environments), thereby limiting cross-client generalization.

E. New-Client Compatibility: A Key Challenge

In P2P-FL, new clients may join the system at arbitrary times, introducing the additional challenge of new-client compatibility. This concept refers to how effectively a trained model can adapt to previously unseen clients within a limited number of gradient update steps. This setting reflects a realistic deployment scenario in which clients dynamically enter the system after the initial training phase.

Achieving strong new-client compatibility is particularly challenging because it requires representations to be both discriminative and transferable. A model that overfits to the training clients may achieve high performance on known clients but fail to generalize to unseen ones. Conversely, a model that learns weak or insufficiently expressive representations may fail to adapt effectively to new clients, even after fine-tuning.

F. Our Approach

To address representation drift and improve new-client compatibility, we preserve transferable knowledge in the representation space, ensuring that discriminative features learned by each client remain stable and reusable across sequential peer-to-peer updates. We mitigate drift by combining supervised contrastive learning [7] with relational knowledge distillation (RKD) [8]. Supervised contrastive learning promotes intra-class compactness and inter-class separation, improving class discriminability, whereas RKD preserves structural relationships by aligning inter-sample distances and angular triplets, thereby capturing scale- and dimension-invariant geometric information. Together, these objectives stabilize manifold topology across updates, reducing drift and improving cross-client generalization and adaptation to new clients in P2P-FL.

G. Contributions

Our key contributions are summarized as follows:

- We identify representation drift as a fundamental limitation of parameter-space regularization in P2P-FL and analyze its impact on cross-client generalization and new-client compatibility.
- We propose a representation-preserving objective that combines supervised contrastive learning with relational knowledge distillation to improve representation learning and preservation across sequential client updates under non-IID data distributions.
- Extensive experiments demonstrate that the proposed approach significantly improves new-client compatibility while maintaining competitive client-level performance, without modifying the FedPC communication protocol or increasing communication overhead.

II. Related Work

A. Knowledge-Distillation-Based Approaches in FL

In client-to-server FL, knowledge distillation (KD) is widely used to mitigate global model performance degradation under non-IID data distributions. Several data-free KD methods have been proposed to preserve privacy while improving generalization. FedGen [9] employs a server-side generator to aggregate client knowledge without proxy data and to guide local training. FedDFKD [10] transfers knowledge between clients and the server using a lightweight delivery model, without requiring public datasets. FedF2DG [11] addresses non-IID settings by generating pseudo-data from client models and adaptively controlling label distributions.

Despite their effectiveness, these approaches rely on centralized server coordination for knowledge distillation and are therefore unsuitable for P2P-FL systems.

B. Federated Learning in NDAR

Recent studies have applied FL to driver state and behavior understanding, addressing privacy and personalization challenges. A personalized FL-based stress monitoring system [12] prioritizes training samples from drivers with similar profiles to improve individual stress estimation. The pFedCI framework [13] enables clients to fine-tune a global model using adaptive cognitive models for personalized driver behavior recognition, while preserving both privacy and global generalization. An FL-based driver activity recognition framework [14] demonstrates that decentralized training across vehicular data sources can achieve performance comparable to centralized approaches, highlighting FL's practicality for privacy-preserving driver activity analysis.

III. Problem Formulation

A. System Model

We consider a peer-to-peer federated learning (P2P-FL) system consisting of a set of N clients $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_N\}$. Each client $\mathcal{C}_i$ owns a private dataset $\mathcal{D}_i = \{(x_{ij}, y_{ij})\}_{j=1}^{|\mathcal{D}_i|}$, drawn from a client-specific data distribution $\mathcal{P}_i$. These distributions are assumed to be non-IID, reflecting heterogeneity in user behavior, data collection environments, and sensing conditions. A shared model is sequentially propagated across clients according to a predefined or randomized gossip order, without requiring a central server.

Let θ_t denote the model parameters after client $\mathcal{C}_t$ updates the received model θ_{t-1} using its local dataset $\mathcal{D}_t$, yielding θ_t. This formulation naturally aligns P2P-FL with a continual learning paradigm, where each client corresponds to a task encountered sequentially.

B. FedPC Objective

FedPC addresses catastrophic forgetting in P2P-FL by introducing a parameter-space regularization term that penalizes deviation between successive model updates. Specifically, the local optimization objective at client $\mathcal{C}_t$ is defined as

$$\mathcal{L}_{\text{FedPC}}(\theta_t) = \mathcal{L}_{\text{CE}}(\theta_t; \mathcal{D}_t) + \frac{\mu}{2}\|\theta_t - \theta_{t-1}\|^2, \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss computed on the local dataset $\mathcal{D}_t$, and $\mu > 0$ controls the strength of the proximal regularization. The proximal term penalizes deviations between the current model parameters and those received from the previous client.

C. Problem Statement

In a P2P-FL system with sequential client updates and highly non-IID data, our goal is to design a local learning objective that preserves transferable knowledge across clients. Specifically, we aim to improve compatibility with new clients while maintaining competitive client-level performance and cross-client generalization.

We seek to achieve this goal under the following constraints: (i) no centralized coordination or aggregation, (ii) no sharing of raw data or auxiliary memory across clients, and (iii) no additional communication overhead beyond the standard peer-to-peer protocol.

IV. Proposed Method

A. Overview of the Proposed Approach

The primary objective of this work is to preserve transferable knowledge in P2P-FL by explicitly constraining the evolution of feature representations across clients. Unlike FedPC, which mitigates catastrophic forgetting by penalizing parameter deviations, our approach operates directly in the representation space. We argue that preserving the geometry of the representation space is more effective for maintaining knowledge that generalizes across clients while remaining adaptable to unseen clients.

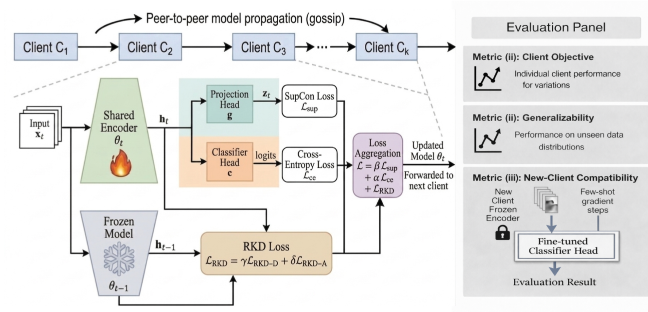

Fig. 2. **Overview of our proposed framework.** Clients sequentially exchange a shared model, while supervised contrastive learning and relational knowledge distillation preserve the geometry of the representation space across clients with non-IID data distributions.

We propose a representation-preserving learning objective that combines two complementary components: supervised contrastive learning and relational knowledge distillation. Supervised contrastive learning promotes the formation of discriminative and well-clustered features within each client, improving representation quality and class separability. Relational knowledge distillation preserves structural relationships among features across successive client updates, thereby mitigating representation drift induced by sequential training on non-IID data.

Our approach does not alter the peer-to-peer communication protocol used in FedPC and introduces no additional communication overhead.

B. Supervised Contrastive Learning

Supervised contrastive learning encourages samples of the same class to form compact clusters in the feature space while pushing samples from different classes apart. The supervised contrastive loss function is shown in Eq. 2, in which z_i and z_p are positive feature pairs belonging to the positive set $P(i)$, and $A(i)$ stands for the negative sets for z_i. τ is the temperature scaling factor that can tune the discrimination between the positive and negative sets.

$$\mathcal{L}_{\text{SupCon}} = -\sum_{i\in I}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp\left(z_i\cdot z_p/\tau\right)}{\sum_{a\in A(i)}\exp\left(z_i\cdot z_a/\tau\right)} \tag{2}$$

By explicitly enforcing class-wise clustering, supervised contrastive learning yields representations that are more robust and transferable than those learned with cross-entropy loss alone.

C. Relational Knowledge Distillation

To preserve the structural topology of the feature space across incremental updates in a P2P-FL setting, we employ relational knowledge distillation (RKD). Unlike logit-based distillation, RKD transfers structural knowledge from a frozen teacher model (from the previous round) to the current student model by penalizing discrepancies in the relative geometric positions of samples.

1) Distance-wise Relational Distillation: Distance-wise relational distillation aims to preserve the global structural manifold by aligning inter-sample Euclidean distances within the feature space. For a set of samples $\mathcal{N}$, let h_i^{t-1} and h_i^t denote the feature representations of sample i from the teacher and student models, respectively. The inter-sample distances are defined as $D_{ij}^{t-1} = \|h_i^{t-1} - h_j^{t-1}\|_2$ for the teacher and $D_{ij}^t = \|h_i^t - h_j^t\|_2$ for the student. To mitigate scale dependency caused by temporal heterogeneity across client updates, we normalize these distances by their mean computed over all distinct sample pairs, i.e., $\mu^{t-1} = \mathbb{E}_{i\neq j}[D_{ij}^{t-1}]$ and $\mu^t = \mathbb{E}_{i\neq j}[D_{ij}^t]$.

The distance-wise relational distillation loss, $\mathcal{L}_{\mathrm{RKD-D}}$, is then formulated as the squared difference between these normalized distances:

$$\mathcal{L}_{\mathrm{RKD-D}} = \sum_{i,j\in\mathcal{N}} \left(\frac{D_{ij}^t}{\mu^t} - \frac{D_{ij}^{t-1}}{\mu^{t-1}} \right)^2 \tag{3}$$

2) Angle-wise Relational Distillation: To capture local geometric structure, we utilize angle-wise relational distillation, which aims to preserve the angular relationships among feature triplets. For a triplet of features (i, j, k), we define the angular similarity ψ_A as the cosine similarity corresponding to the angle at vertex j. Let $e_{ij} = \frac{h_i - h_j}{\|h_i - h_j\|_2}$ and $e_{kj} = \frac{h_k - h_j}{\|h_k - h_j\|_2}$ denote the unit vectors originating from h_j toward h_i and h_k, respectively. The similarity is then computed as: $\psi_A(h_i, h_j, h_k) = \langle e_{ij}, e_{kj} \rangle$.

The angle-wise relational distillation loss, $\mathcal{L}_{\mathrm{RKD-A}}$, minimizes the squared difference between the scalar angular similarities produced by the teacher and student models across all distinct triplets:

$$\mathcal{L}_{\mathrm{RKD-A}} = \sum_{\substack{i,j,k\in\mathcal{N} \\ i\neq j\neq k}} \left(\psi_A(h_i^{t-1}, h_j^{t-1}, h_k^{t-1}) - \psi_A(h_i^t, h_j^t, h_k^t) \right)^2 \tag{4}$$

D. Final Training Objective

For each client $\mathcal{C}_t$, the final training objective is a weighted combination of the cross-entropy loss, the supervised contrastive loss, and relational knowledge distillation loss terms.

$$\mathcal{L} = \alpha\mathcal{L}_{\text{CE}} + \beta\mathcal{L}_{\text{SupCon}} + \gamma\mathcal{L}_{\text{RKD-D}} + \delta\mathcal{L}_{\text{RKD-A}} \tag{5}$$

where α, β, γ, and δ control the relative contribution of each component.

E. Peer-to-Peer Communication Protocol

Gossip-based protocols [15] are widely used in distributed systems due to their scalability and robustness, enabling efficient information dissemination through randomized local interactions. Our method adopts the peer-to-peer gossip protocol of FedPC [5], where clients are organized in a randomized gossip order and sequentially perform local training. After each local update, the model is forwarded to the next client in the communication chain for further refinement. The approach incurs no additional communication overhead and requires no exchange of auxiliary information between clients.

V. Experimental Setup

A. Datasets

We evaluate the proposed method on vision-based datasets for naturalistic driving action recognition, specifically, the **StateFarm** dataset [16] and the **AICity** dataset [17], [18]. Both datasets consist of in-vehicle camera images capturing a variety of driver activities, including safe driving, texting, talking on the phone, and other potentially distracting behaviors. Each image is annotated with both a driver activity label and a driver identity. In our federated learning setup, each driver is treated as an individual client, and the datasets exhibit pronounced heterogeneity across clients, capturing variations in driving behavior, style, and environmental conditions.

B. Baselines

We compare the proposed method against several baselines to evaluate its effectiveness:

- **Independent Training**: Each client trains a model using only its local data, without any form of model sharing. This baseline represents a lower bound on performance when neither data nor model updates are shared across clients.
- **FedAvg** [2]: A centralized federated learning method that aggregates client updates using weighted averaging.
- **FedProx** [19]: A centralized federated learning method that incorporates a proximal term to address data heterogeneity.
- **FedPC** [5]: A peer-to-peer federated learning baseline that formulates training as a continual learning problem with parameter-space proximal regularization.

C. Ablation Studies

To isolate the contribution of each component of the proposed method, we conduct a series of ablation studies:

- **CE only**: Training using only the cross-entropy loss, without any representation-level regularization.
- **CE + SupCon (w/o RKD)**: Training using the cross-entropy and supervised contrastive losses, without the relational knowledge distillation loss.
- **CE + RKD (w/o SupCon)**: Training using the cross-entropy and relational knowledge distillation losses, without the supervised contrastive loss.
- **Full Objective (Ours)**: Training using the full objective, combining the cross-entropy, supervised contrastive, and relational knowledge distillation losses.

These ablations enable us to evaluate the individual and combined effects of representation formation and preservation in peer-to-peer federated learning.

D. Implementation Details and Evaluation Protocol

Following [5], we utilize an ImageNet-pretrained ResNet-34 [20] as the shared feature encoder, with the first three blocks frozen during training. The projection head is a Multi-Layer Perceptron (MLP) with one hidden layer (ReLU) and an output size of 128, followed by ℓ_2 normalization. The classifier head comprises a single fully connected layer.

The full set of clients is partitioned into two disjoint subsets, with 80% assigned to training clients and 20% to held-out clients. Training clients participate in P2P-FL, while held-out clients are reserved for evaluating new-client compatibility. No data from held-out clients is used during training. However, during evaluation, one or more gradient descent steps may be performed to personalize the model on held-out clients. This partitioning ensures that evaluation is conducted on data unseen during training. For each training client, the local dataset is further divided into training and test sets using an 80:20 split. Local training is performed using the Adam optimizer with an initial learning rate of 10^{-4}, which is decayed by a factor of 0.5 at each communication round. We use a weight decay of 10^{-5} and a batch size of 128 for all experiments. All images are resized to 224×224 to reduce storage and computational overhead. Peer-to-peer training is conducted for five communication rounds, each consisting of five local epochs. Clients are selected and ordered according to a randomized gossip strategy in each round. The hyperparameters $\alpha, \beta, \gamma,$ and δ are set to $0.1, 0.5, 1.0,$ and 1.0, respectively. The temperature parameter τ is set to 0.07.

We follow the FedPC [5] evaluation protocol and report three metrics: client objective, generalizability, and new-client compatibility. The client objective is computed by evaluating the final model on the local test set of each training client. Generalizability is assessed by evaluating a model trained on one client across the test sets of other training clients and averaging the resulting performance. For new-client compatibility, adaptation is performed using SGD with a learning rate of 10^{-3} for a small number of gradient update steps. All experiments are conducted on a single NVIDIA Tesla T4 GPU.

VI. Results and Analysis

A. Overall Performance Trends

We evaluate the proposed method using the three metrics introduced in FedPC on two datasets: the StateFarm dataset and the AICity dataset. Figures 3 and 4 report performance across peer-to-peer training rounds for all methods.

Across both datasets, the proposed approach consistently achieves strong performance across all three metrics. Metric (i) measures classification accuracy on the local test sets of participating clients. As shown in Figures 3(a) and 4(a), performance on the client objective improves steadily over training rounds for all peer-to-peer methods.

Metric (ii) evaluates cross-client generalization by measuring performance on the test data of other clients. Figures 3(b) and 4(b) show that the proposed method consistently outperforms both centralized and peer-to-peer baselines across rounds. While Federated Averaging and FedProx exhibit gradual improvements, their performance remains limited under non-IID client data distributions.

Metric (iii) evaluates adaptability to previously unseen clients under a few-step adaptation setting. As shown in Figures 3(c) and 4(c), the proposed method achieves substantially higher adaptation performance than all baselines on both datasets, especially with fewer gradient update steps.

B. Representation Drift Analysis

To analyze representation drift in the P2P-FL setting, we quantify it using two complementary metrics (see Eqs. 4 and 3). Local angular drift (Fig. 5(a)) captures higher-order structural changes by measuring the deviation in geometric orientations among feature triplets. In contrast, global relational drift (Fig. 5(b)) quantifies distortions in the feature space manifold by measuring shifts in the normalized pairwise distance matrix. Together, these metrics provide a comprehensive assessment of representation stability across sequential peer updates. As shown in Figure 5, the proposed method consistently reduces both local angular and global relational drift compared to FedPC, accounting for its superior feature transferability and enhanced compatibility with incoming clients.

VII. Conclusion and Future Work

In this work, we propose a representation-preserving learning objective for peer-to-peer federated learning that combines supervised contrastive loss with relational knowledge distillation. By operating directly in the representation space, the proposed method addresses a key limitation of parameter-space regularization approaches such as FedPC. Extensive experiments demonstrate consistent improvements in cross-client generalization and substantial gains in adaptation to new clients.

Future work includes adaptive scheduling of representation-level constraints to balance generalization and specialization, scaling to larger federated systems, and developing theoretical guarantees for representation stability and generalization.

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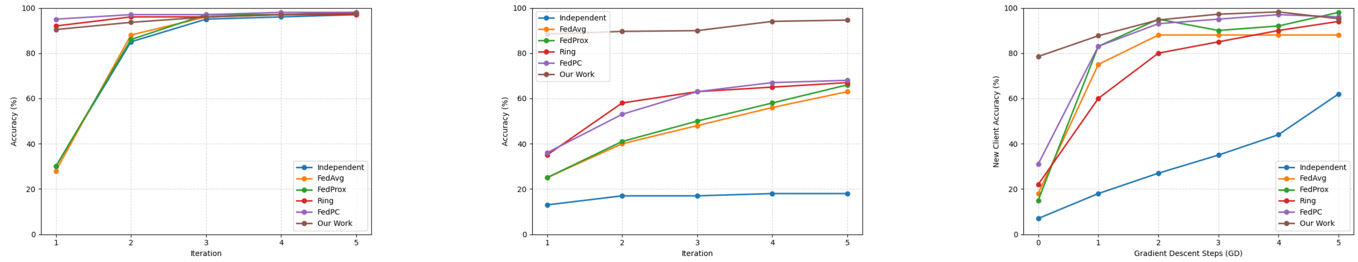
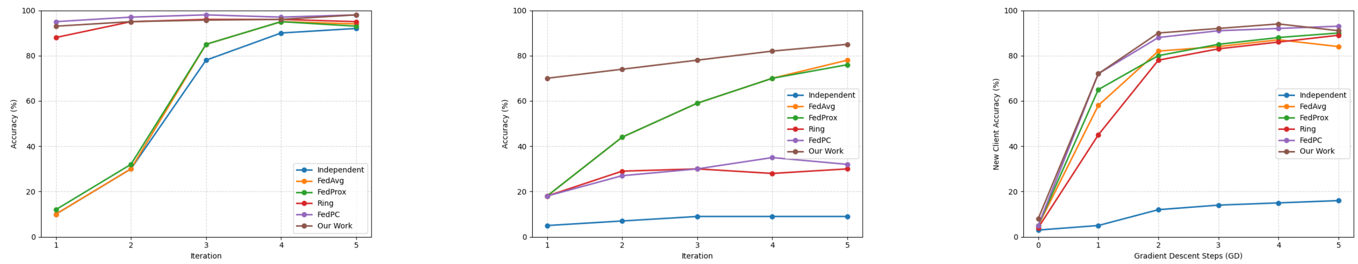


(a) Client Objective (Metric i) (b) Generalizability (Metric ii) (c) New-Client Compatibility (Metric iii)

Fig. 3. Performance of different methods on the **StateFarm** dataset under the three FedPC evaluation metrics. Results are reported on held-out test sets, with each data point representing the average performance across all clients.



(a) Client Objective (Metric i) (b) Generalizability (Metric ii) (c) New-Client Compatibility (Metric iii)

Fig. 4. Performance of different methods on the **AICity** dataset under the three FedPC evaluation metrics. Results are reported on held-out test sets, with each data point representing the average performance across all clients.

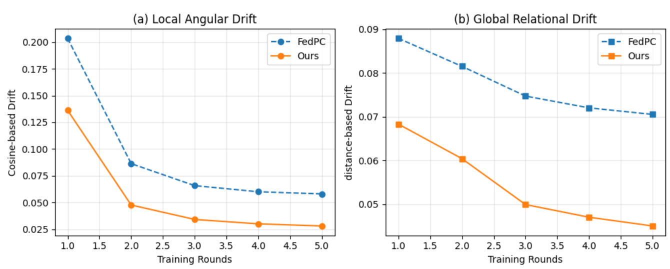


Fig. 5. Comparison of representation drift across peer-to-peer training rounds. (a) Local angular drift: Quantifies shifts in the geometric orientation of feature triplets. (b) Global relational drift: Quantifies distortions in the pairwise distance structure across the feature manifold. Lower values signify enhanced representation stability.

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